

Trichilemmal carcinoma of the scalp.

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This report describes the case of an 86-year-old male who presented with a large scalp tumour. Imaging revealed a large vertex mass, without intracranial extension, and multiple nodular subcapsular hepatic lesions suspected to be secondary in nature. Surgical resection was performed. Pathological examination revealed an adnexal carcinoma of follicular origin, thus a trichilemmal carcinoma. Controlled wound healing (budding of the diploë) was completely successful within 12 months. The patient refused the assessment and treatment of his metastases.

The experience of our Clinic in laparoscopy for adnexal masses and the correlation between ultrasound findings and pathological results.

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An adnexal mass (mass of the ovary, Fallopian tube, or surrounding connective tissues) is a common gynecological problem. Ovarian pathology can occur at any time from fetal life to menopause. Sonography is a clinically important imaging modality for assessing whether an adnexal mass is likely benign or possibly malignant. Most ovarian surgeries are for benign disease and can be performed laparoscopically. We retrospectively evaluated all the patients from our Clinic who underwent laparoscopy for adnexal tumors in the last three year. We were studying the correlation of the preoperative ultrasound examination and the pathological result for each case. In this study, there were included 71 patients who underwent a laparoscopic intervention for adnexal tumors. The average age was 33 years old (range 18 and 60 years old). The ultrasound findings were ovarian endometriosis in 23.9% of cases, left ovarian cysts (22.5%) and right ovarian tumors (18.3%), and the lowest percentage was of left ovarian tumors (8.5%). Regarding the histopathological examination, we found the following results: ovarian endometriosis (32.4%), ovarian functional cysts (15.5%), serous ovarian cysts (15.5%), salpingitis (12.7%), ovarian myoma (7%), papillary serous ovarian cysts (4.2%), ovarian teratoma cysts (9.9%), one case (1.4%) of ovarian carcinoma and one case (1.4%) of borderline serous tumor. In this study, we observed that it was a strong correlation between the ultrasound finding and the pathological result for adnexal tumors ($p < 0.001$). The laparoscopy was performed in our Clinic for adnexal masses with benign characters in ultrasound examination. The histopathological examination confirmed the diagnosis, being only one case of ovarian carcinoma in this study.

Adnexal masses caused by pelvic inflammatory disease: MR appearance.

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To describe the morphologic and signal intensity characteristics of inflammatory adnexal masses in magnetic resonance (MR) images. MR images of 15 patients with a total of 20 confirmed inflammatory adnexal masses were analyzed retrospectively. The findings obtained from MR imaging were as follows: lesions have ill-defined borders; ill-defined areas of high intensity signals surround the masses on T(2)-weighted images; a "rim" at the innermost layer of the cyst wall exhibits high signal intensity on T(1)-weighted images and low intensity on T(2)-weighted images; secondary changes in the surrounding fat plane and bowel loops are extensive; and adhesions are dense. These findings reflect the basic

characteristics of the inflammatory process, which are edema caused by increased permeability and highly vascularized granulation tissue caused by the strong tendency to repair. MR imaging findings well reflect the basic pathological process of inflammation, and inflammatory adnexal masses are associated with relatively specific appearances in MR images. MR imaging can be a problem-solving alternative in the diagnosis of inflammatory adnexal masses.

Adnexal masses and malignancies of importance to the colorectal surgeon.

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In this article, the authors review both benign and malignant ovarian masses, as the colorectal surgeon who encounters an adnexal mass at the time of surgery should be aware of the steps necessary for surgical staging and optimal tumor resection. Ovarian tumors—most of which are benign—are divided into three major categories, in order of frequency: epithelial, germ cell, and sex cord-stromal tumors. Nonneoplastic conditions of the ovary that may present as adnexal masses include the following, according to World Health Organization (WHO) classification: pregnancy luteoma, hyperplasia of ovarian stroma, hyperthecosis, massive edema, solitary follicle cysts and corpus luteal cysts, multiple follicle cysts, and endometriosis. Epithelial ovarian tumors arise from the surface epithelium and can be benign or malignant. Histologic types are serous, mucinous, endometrioid, clear cell, or Brenner. Germ cell tumors are more likely to appear in females under 20 years, accounting for 70% of ovarian tumors in this age group. Approximately 3% are malignant. Teratomas are the most common germ cell tumors. Malignancies, in addition to malignant teratomas, include dysgerminomas, endodermal sinus tumors, and embryonal carcinomas. The more common sex cord-stromal tumors include granulosa stromal cell tumors, Sertoli-Leydig cell tumors, and gynandroblastomas. Surgical staging and optimal tumor resection are also addressed, with a focus on epithelial malignancies, as they are the most relevant to colorectal surgeons.

Magnetic resonance imaging of the female pelvis.

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Evaluation of pelvic pathology is an important part of most radiology practices. Magnetic resonance imaging (MRI) has proven itself to be a useful and cost-effective method in evaluating many of these diseases. This article reviews appropriate MR imaging techniques and findings of common gynecologic disorders. Uterine pathology is categorized into congenital anomalies, benign lesions (leiomyomas, adenomyosis, endometrial polyps, and nabothian cysts), and malignancies (endometrial and cervical carcinoma). Adnexal pathology reviewed includes endometriosis, polycystic ovaries, teratomas, and benign and malignant ovarian epithelial tumors.

Guideline No. 403: Initial Investigation and Management of Adnexal Masses.

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To aid primary care physicians, emergency medicine physicians, and gynaecologists in the initial

investigation of adnexal masses, defined as lumps that appear near the uterus or in or around ovaries, fallopian tubes, or surrounding connective tissue, and to outline recommendations for identifying women who would benefit from a referral to a gynaecologic oncologist for further management. Gynaecologists, obstetricians, family physicians, general surgeons, emergency medicine specialists, radiologists, sonographers, nurses, medical learners, residents, and fellows. Adult women 18 years of age and older presenting for the evaluation of an adnexal mass. Women with adnexal masses should be assessed for personal risk factors, history, and physical findings. Initial evaluation should also include imaging and laboratory testing to triage women for management of their care either by a gynaecologic oncologist or as per SOGC guideline no. 404 on the initial investigation and management of benign ovarian masses. A search of PubMed, Cochrane Wiley, and the Cochrane systematic reviews was conducted in January 2018 for English-language materials involving human subjects published since 2000 using three sets of terms: (i) ovarian cancer, ovarian carcinoma, adnexal disease, ovarian neoplasm, adnexal mass, fallopian tube disease, fallopian tube neoplasm, ovarian cyst, and ovarian tumour; (ii) the above terms in combination with predict neoplasm staging, follow-up, and staging; and (iii) the above two sets of terms in combination with ultrasound, tumour marker, CA 125, CEA, CA19-9, HE4, multivariable-index-assay, risk-of-ovarian-malignancy-algorithm, risk-of-malignancy-index, diagnostic imaging, CT, MRI, and PET. Relevant evidence was selected for inclusion in descending order of quality of evidence as follows: meta-analyses, systematic reviews, guidelines, randomized controlled trials, prospective cohort studies, observational studies, non-systematic reviews, case series, and reports. Additional articles were identified through cross-referencing the identified reviews. The total number of studies identified was 2350, with 59 being included in this review. The content and recommendations were drafted and agreed upon by the authors. The Executive and Board of the Society of Gynecologic Oncology of Canada reviewed the content and submitted comments for consideration. The Board of Directors of the Society of Obstetricians and Gynaecologists of Canada approved the final draft for publication. The quality of evidence was rated using the criteria described in the Grading of Recommendations Assessment, Development, and Evaluation (GRADE) methodology framework (Table A1 of Online Appendix A). See Table A2 of Online Appendix A for the interpretation of strong and weak recommendations. The summary of findings is available upon request. Adnexal masses are common, and guidelines on how to triage them and manage the care of patients presenting with adnexal masses will continue to guide the practice of primary care providers and gynaecologists. Ovarian cancer outcomes are improved when initial surgery is performed by a gynaecologic oncologist, likely as a result of complete surgical staging and optimal cytoreduction. Given these superior outcomes, guidelines to assist in the triage of adnexal masses and the referral and management of the care of patients with an adnexal mass are critical. RECOMMENDATIONS (GRADE RATINGS IN PARENTHESES).

Myasthenia gravis genome-wide association study implicates AGRN as a risk locus.

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Myasthenia gravis (MG) is a rare autoimmune disorder affecting the neuromuscular junction (NMJ). Here, we investigate the genetic architecture of MG via a genome-wide association study (GWAS) of the largest MG data set analysed to date. We performed GWAS meta-analysis integrating three different data sets (total of 1401 cases and 3508 controls). We carried out human leucocyte antigen (HLA) fine-mapping, gene-based and tissue enrichment analyses and investigated genetic correlation with 13 other autoimmune disorders as well as pleiotropy across MG and correlated disorders. We confirmed the previously reported MG association with TNFRSF11A (rs4369774; $p=1.09 \times 10^{-13}$, OR=1.4). Furthermore, gene-based analysis revealed AGRN as a novel MG susceptibility gene. HLA fine-mapping pointed to two

independent MG loci: HLA-DRB1 and HLA-B. MG onset-specific analysis reveals differences in the genetic architecture of early-onset MG (EOMG) versus late-onset MG (LOMG). Furthermore, we find MG to be genetically correlated with type 1 diabetes (T1D), rheumatoid arthritis (RA), late-onset vitiligo and autoimmune thyroid disease (ATD). Cross-disorder meta-analysis reveals multiple risk loci that appear pleiotropic across MG and correlated disorders. Our gene-based analysis identifies AGRN as a novel MG susceptibility gene, implicating for the first time a locus encoding a protein (agrin) that is directly relevant to NMJ activation. Mutations in AGRN have been found to underlie congenital myasthenic syndrome. Our results are also consistent with previous studies highlighting the role of HLA and TNFRSF11A in MG aetiology and the different risk genes in EOMG versus LOMG. Finally, we uncover the genetic correlation of MG with T1D, RA, ATD and late-onset vitiligo, pointing to shared underlying genetic mechanisms.

Aptamer-functionalized nucleic acid nanotechnology for biosensing, bioimaging and cancer therapy.

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Nucleic acids have enabled the fabrication of self-assemblies and dynamic operations. Among different functional nucleic acids, aptamers can specifically bind to a wide range of targets, including proteins, viral antigens, living cells and even tissues, and have thus emerged as molecular recognition tools in molecular medicine. Hence, aptamer-functionalized nucleic acid nanotechnology offers applications of biosensing, bioimaging, and cancer therapy. In this review, after a brief overview of nucleic acid nanotechnology, we focus on the integration of aptamers with nucleic acid nanotechnology, including self-assembly constructions and dynamic molecular manipulations. The emerging applications in molecular medicine are subsequently reviewed with aptamer-based self-assemblies and aptamer-involved dynamic molecular manipulation. For convenience, applications are broadly categorized into biosensing, bioimaging, and cancer therapy. Finally, challenges and potential development of nucleic acid nanotechnology are discussed.

Aptamer Screening: Current Methods and Future Trend towards Non-SELEX Approach.

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Aptamers are nucleic acid sequences that specifically bind with target molecules and are vital to applications such as biosensing, drug development, disease diagnostics, etc. The traditional selection procedure of aptamers is based on the Systematic Evolution of Ligands by an Exponential Enrichment (SELEX) process, which relies on repeating cycles of screening and amplification. With the rapid development of aptamer applications, RNA and XNA aptamers draw more attention than before. But their selection is troublesome due to the necessary reverse transcription and transcription process (RNA) or low efficiency and accuracy of enzymes for amplification (XNA). In light of this, we review the recent advances in aptamer selection methods and give an outlook on future development in a non-SELEX approach, which simplifies the procedure and reduces the experimental costs. We first provide an overview of the traditional SELEX methods mostly designed for screening DNA aptamers to introduce the common tools and methods. Then a section on the current screening methods for RNA and XNA is prepared to demonstrate the efforts put into screening these aptamers and the current difficulties. We

further predict that the future trend of aptamer selection lies in non-SELEX methods that do not require nucleic acid amplification. We divide non-SELEX methods into an immobilized format and non-immobilized format and discuss how high-resolution partitioning methods could facilitate the further improvement of selection efficiency and accuracy.

An overview of aptamer: Design strategy, prominent applications, and potential challenge in plants.

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Aptamers, serving as highly efficient molecular recognition and biotechnology tools, have garnered increasing interest in the realm of plant science in recent years. Aptamers are synthetic single-stranded short nucleotides or peptides, that bind targets with high specificity and affinity, triggering precise biological responses. As an alternative to antibodies, aptamers present promising avenues for advancement in biological researches. Aptamers function in a range of fields, encompassing cell signaling, drug development, biosensor technology, as well as botany, agricultural and forestry sciences. In this review, we introduce classifications and screening methods of aptamers, as well as aptamer-based technologies, highlighting their significant contributions to recent advancements. With their powerful functionality and ability to bind targets with high specificity and affinity, aptamers offer promising opportunities for breakthroughs in plant research.
